

60.4 on AIME 2024

Surpassing DAPO by 10+ points on the same Qwen2.5-32B model (no SFT data)

Authors: ByteDance Seed
Framework: Value-model-based RL
Task: Long chain-of-thought reasoning

Headline Performance Snapshot

Method	AIME 2024
Vanilla PPO	5
DeepSeek-R1-Zero-Qwen-32B	47
DAPO	50
VAPO	60.4

Why Value Models? Finer Credit Assignment for Long-CoT Reasoning

Trajectory-level averages miss token-level difficulty variation in multi-step math reasoning.

Value-Model-Free (e.g., GRPO, DAPO)

- Uses trajectory/group-level rewards
- Uniformly applies advantage across response
- Coarser credit assignment for long reasoning chains
- Can under-utilize token-level learning signal



Value-Model-Based (VAPO)

- Learns value estimates along token positions
- Enables token-level credit assignment
- Better suited to long, branching CoT traces
- Outperforms value-free baselines when stabilized

Source: VAPO brief (ByteDance Seed), AIME 2024 results

Three Challenges in Value-Model-Based RL for Long-CoT

VAPO targets failure modes that previously limited value-model approaches.

1) Value Model Bias

Long trajectories + bootstrapping induce biased value estimates, destabilizing policy updates.

2) Heterogeneous Lengths

Short and long responses require different bias-variance trade-offs for robust advantage estimation.

3) Reward Sparsity

Outcome-only rewards are sparse; long CoT exacerbates delayed credit assignment.

Source: VAPO brief (ByteDance Seed), AIME 2024 results

VAPO Framework: Techniques Mapped to Core Challenges

Integrated design: pretraining + adaptive estimation + robust optimization.

Challenges

Value Bias

Length Heterogeneity

Reward Sparsity

VAPO Components

Value-Pretraining

Mitigates early value bias; +49 points contribution in ablation

Decoupled-GAE

Reduces coupling-induced instability in advantage estimation

Length-adaptive GAE

Dynamic λ by response length to optimize bias-variance

Clip-Higher + Token-level Loss

Stabilizes policy updates for long responses

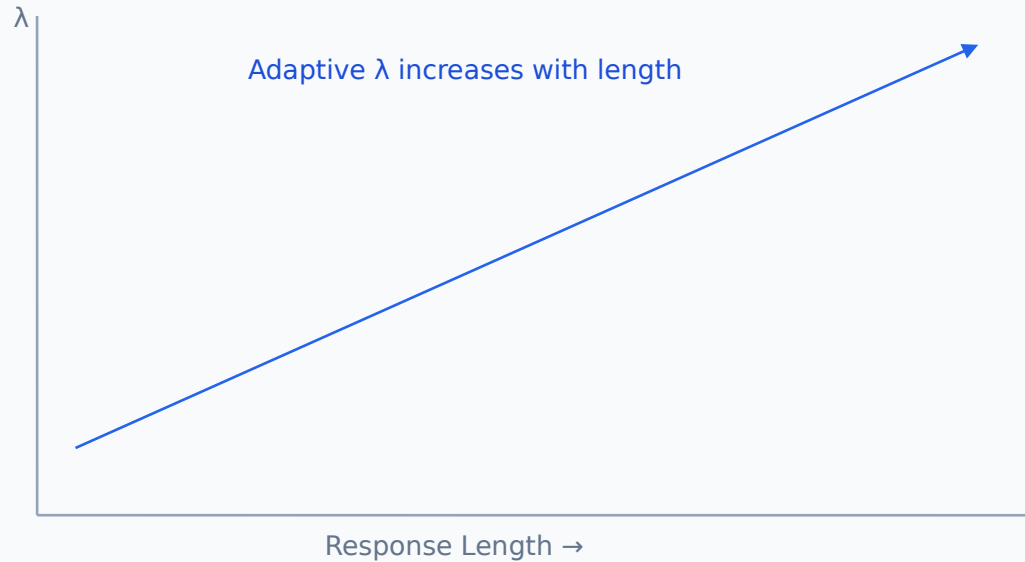
Positive Example LM Loss + Group-Sampling

Improves reward density and learning signal quality

Source: VAPO brief (ByteDance Seed), AIME 2024 results

Length-Adaptive GAE: Dynamic λ for Variable-Length Responses

Short responses prefer lower variance; long responses need lower bias.



Short responses → lower λ

Prioritize lower variance in advantage estimates

Long responses → higher λ

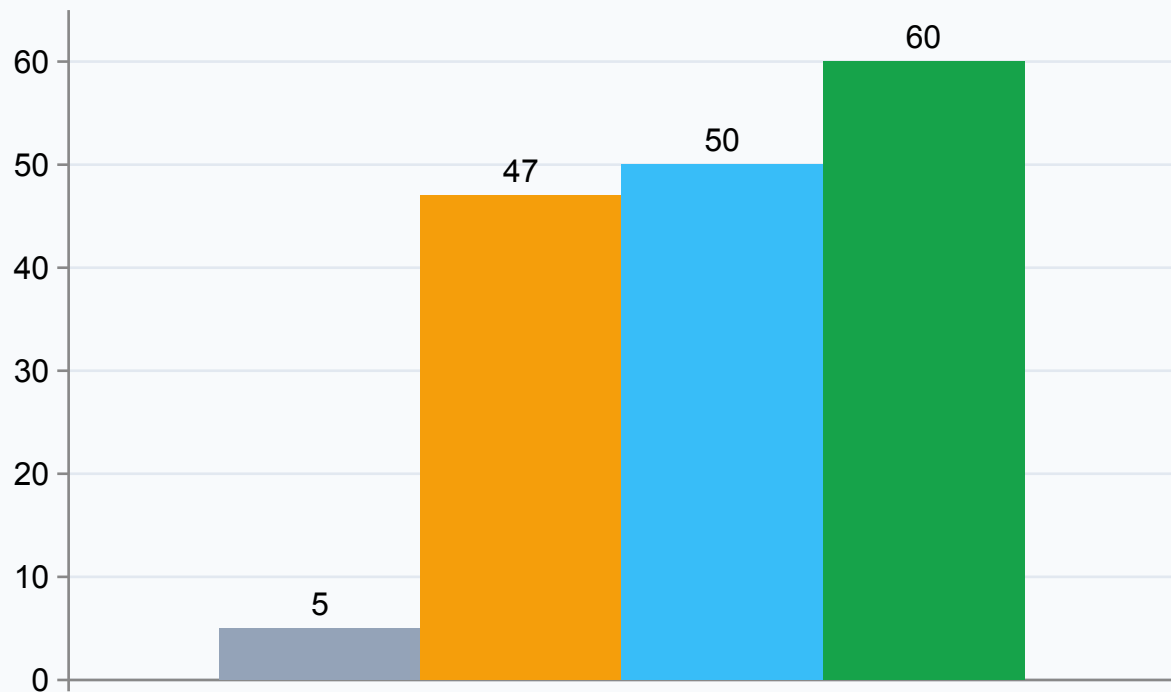
Prioritize lower bias over long-horizon credit assignment

Result: better bias-variance trade-off across heterogeneous sequence lengths

Source: VAPO brief (ByteDance Seed), AIME 2024 results

Main Results: VAPO Leads AIME 2024 by 10+ Points

All compared on Qwen2.5-32B base model; no SFT data for VAPO.



VAPO reaches ~60% by step 5,000, DAPO reaches ~50% by step 5,500.

Source: VAPO brief (ByteDance Seed), AIME 2024 results

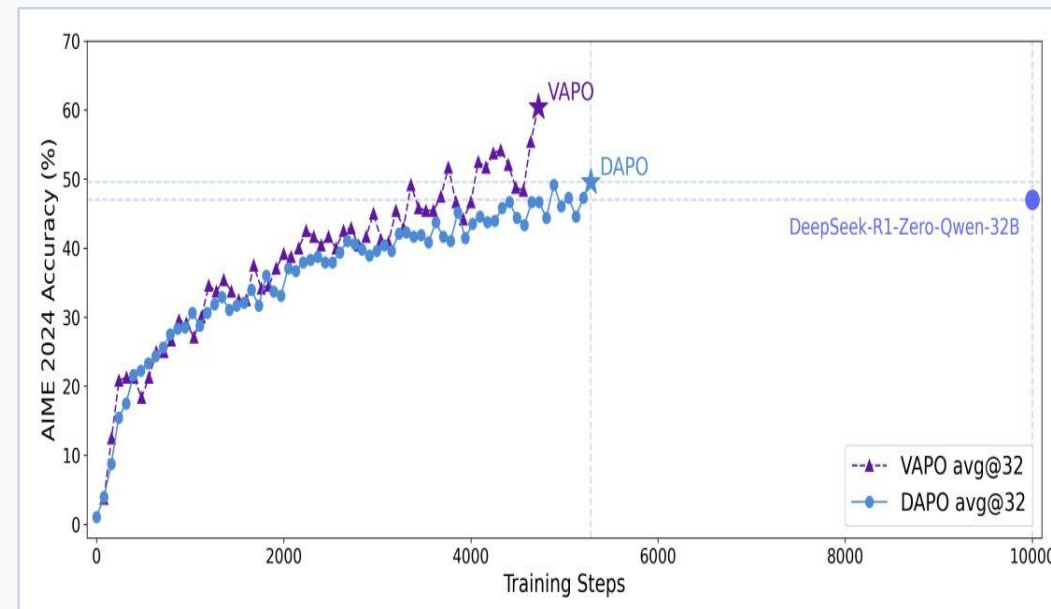


Figure 2: Training accuracy curve (VAPO vs DAPO)

Training Efficiency: SOTA in 5K Steps with Greater Reasoning Length

VAPO is both more sample-efficient and behaviorally distinct from DAPO.

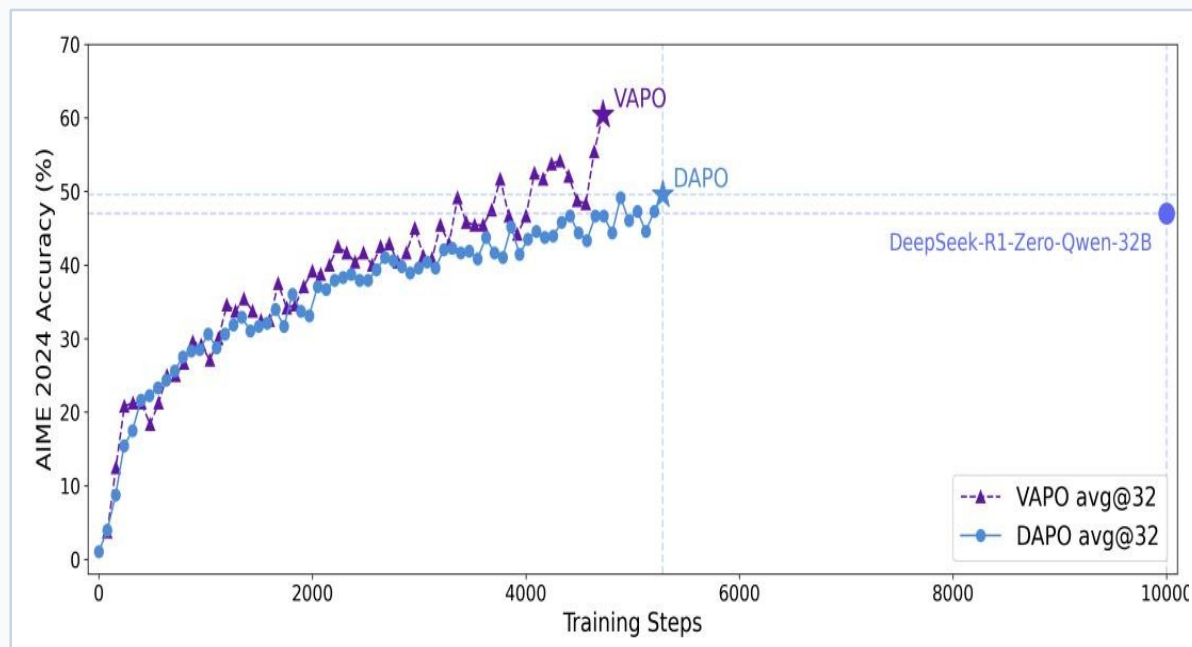


Figure 2: Mean accuracy over training steps

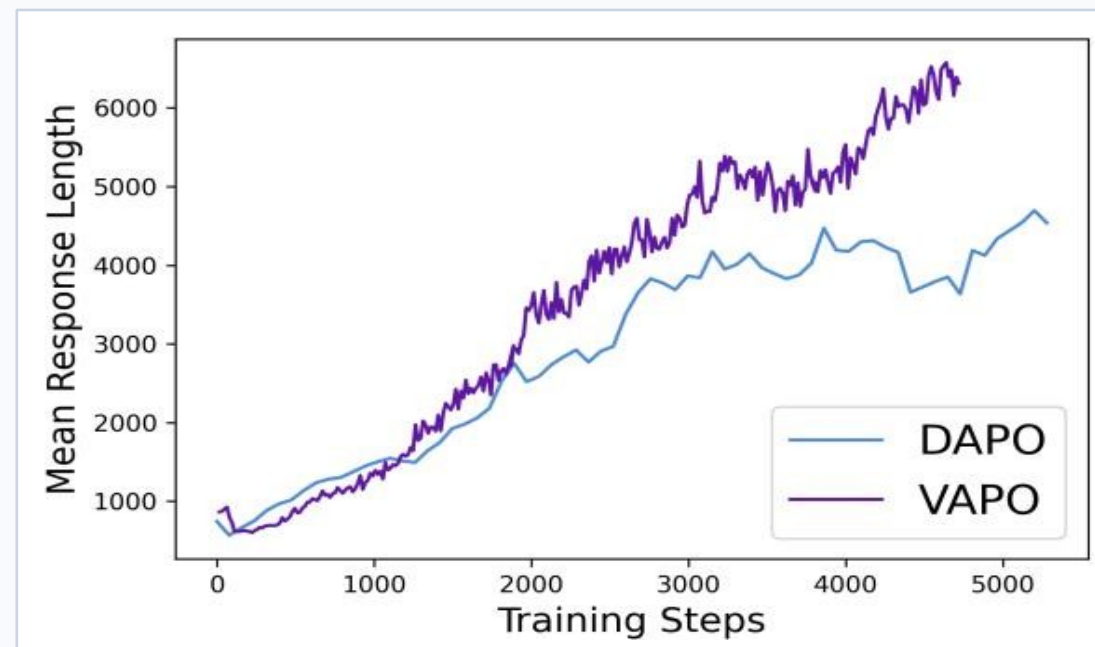


Figure 3: Mean response length over training

VAPO hits 60.4 at ~5K steps; DAPO needs ~10K to reach 50. VAPO also trends to longer responses (~6K vs ~4.5K tokens).

Source: VAPO brief (ByteDance Seed), AIME 2024 results

Ablation: Component Contributions to VAPO Performance

Removing key components sharply degrades AIME 2024 performance.

Configuration	AIME 2024	Δ vs Full VAPO (60)
Full VAPO	60	0
w/o Value-Pretraining	11	-49
w/o Decoupled-GAE	33	-27
w/o Length-adaptive GAE	45	-15

**Largest drop:
Value-Pretraining**

**60 → 11
(−49 points)**

Critical enabler for value-model-based RL

All three components are material; none are redundant for top performance.

Source: VAPO brief (ByteDance Seed), AIME 2024 results

Key Takeaways & Implications for Future Reasoning Models

Value-model-based RL is competitive again when engineered for long-CoT settings.

- VAPO achieves 60.4 on AIME 2024, exceeding DAPO (50) and DeepSeek-R1-Zero-Qwen-32B (47).
- Value-model-based RL can outperform value-free methods when bias, length heterogeneity, and reward sparsity are jointly addressed.
- Value-Pretraining is the highest-leverage component (−49 points when removed).
- Length-adaptive GAE is a key algorithmic advance for heterogeneous response lengths (−15 points when removed).
- VAPO is training-stable (zero crashes across multiple runs) and reaches SOTA in ~5K steps.

Implication: Future reasoning RL should treat value estimation, credit assignment, and training stability as a co-designed system.

Source: VAPO brief (ByteDance Seed), AIME 2024 results